

Meningitis and Meningo-encephalitis

Types, Epidemiology, Etiology
Clinical Features, Diagnosis, Complications,
Management, Prognosis and Prevention

Meningitis and Meningo-encephalitis

Definition, Types

- **Meningitis** is inflammation of membranes surrounding the brain and spinal cord.
- It is usually bacterial in origin.
- **Pyogenic (Bacterial) Meningitis** is due to bacterial infection
- **Tuberculous Meningitis** is caused by *Mycobacterium tuberculosis*
- **Meningo-encephalitis** is inflammation of meninges and the brain matter.
- It is commonly caused by a viral infection

Meningitis and Meningo-encephalitis

Epidemiology

- **CNS infections** caused by bacteria and viruses are more common in infancy and childhood
- **Acute Bacterial and Viral Meningitis** in children has decreased in recent years after immunization of children against common bacterial and viral organisms infecting the CNS
- **Viral Meningo-encephalitis** caused by viruses against which no vaccine is available, still remains common in all countries

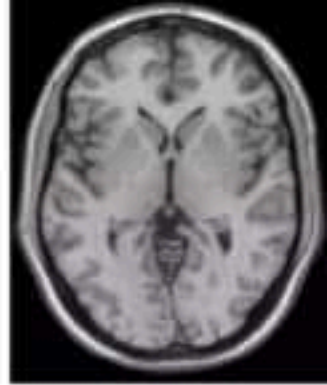


Acute Bacterial Meningitis (Pyogenic Meningitis)

Etiology, Clinical Features, Diagnosis, Complications,
Management, Prognosis and Prevention

Acute Bacterial Meningitis

Etiology



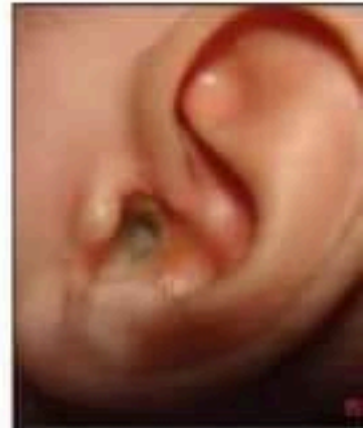
- **Newborns:** E. Coli, Klebsiella, Gram – ve organisms
- **Infants:** Streptococcus pneumoniae, H. influenza, Staph. Aureus
- **Childhood:** Streptococcus pneumoniae, H. influenza, Neisseria meningitidis



Acute Bacterial Meningitis

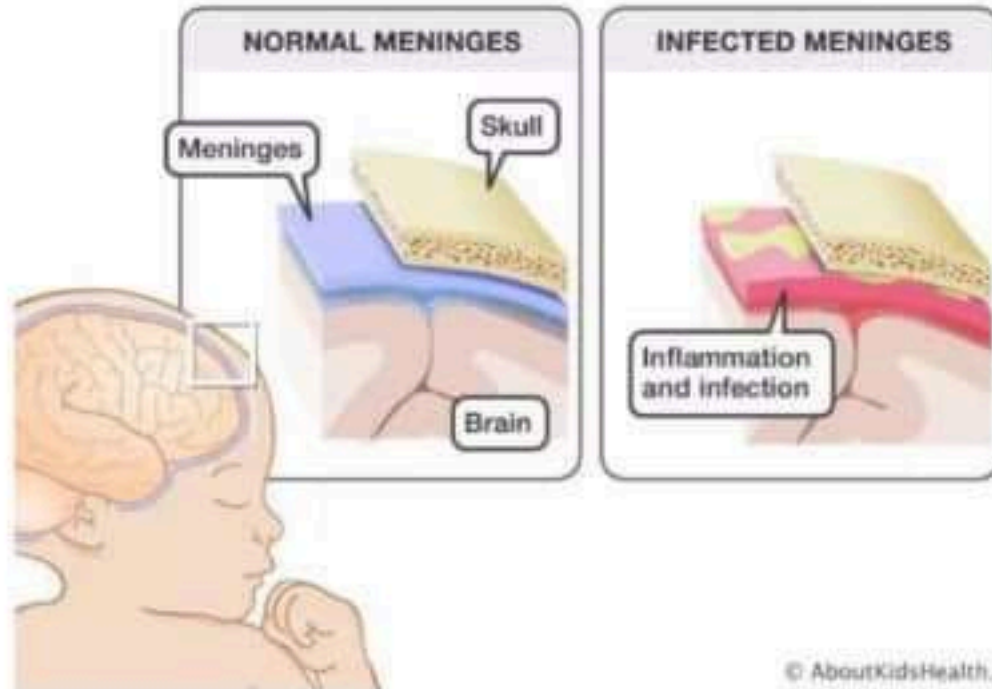
Risk of Infection

- Unvaccinated children
- Immunodeficiency (Nephrotic Syndrome)
- Nasopharyngeal carriage of pathogenic bacteria
- Chronic Otitis Media
- Meningomyelocele



Acute Bacterial Meningitis Pathogenesis

- **Meninges:** Inflammation
- **CSF:** Exudate
- **Brain:** Hypoxia, Ischemia





Acute Bacterial Meningitis (Pyogenic Meningitis)

Clinical Presentation, Diagnosis, Complications

Case Scenario

- A 2 year old child presents with fever since yesterday. Fever has been high grade and child has been crying continuously.
- Today, he had a tonic-clonic seizure lasting 5 minutes in the morning.
- After the seizure child has remained drowsy and has not taken any feed
- On examination, child is febrile with temperature of 102.5 F
- His GCS is 11 and the body has increased tone.
- Neck stiffness is present. Anterior fontanel is closed
- On questioning, parents say that child has missed his vaccinations during infancy
- What is your most likely diagnosis ? Pyogenic Meningitis

Acute Bacterial Meningitis

Clinical Features

- **Symptoms**

- Fever
- Irritability, lethargy
- Headache (in older children), Vomiting
- Seizures / Convulsions

- **Signs**

- Impaired conscious level
- Bulging anterior fontanel
- SOMI (Neck stiffness, Kernig sign, Brudzinski sign)
- Petechiae on skin (meningococcal infection)



Acute Bacterial Meningitis

Evaluation

- **History:**

- Present illness – fever, excessive crying, seizures
- Past History – vaccinations
- Development history – milestones
- Family History – seizures

- **Physical Examination:**

- General Physical Examination – focus of infection
- Neurological Examination – abnormal findings, SOMI
- Direct Ophthalmoscopy – Papilledema (raised ICP)

Acute Bacterial Meningitis

Investigations

- **CSF examination:**

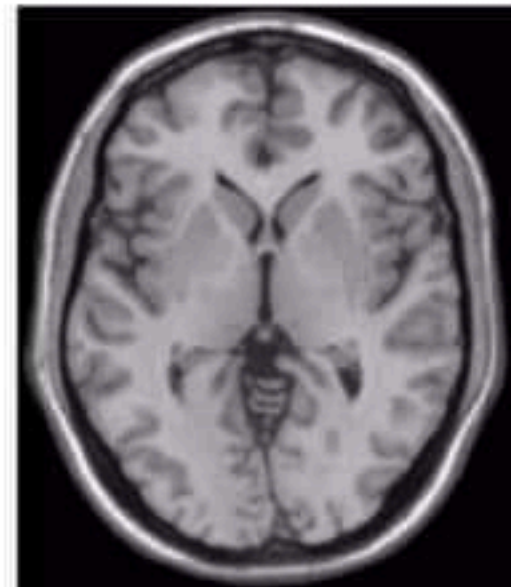
- Biochemistry, Cells
- Gram stain, Bacterial latex agglutination / ELISA / PCR
- Bacterial Culture

- **Blood:**

- CBC, CRP,
- Blood Culture

- **Radiology:**

- CT / MRI Brain



Acute Bacterial Meningitis

Investigations – CSF analysis

CSF analysis in different types of meningitis

	Bact. Meningitis	Viral Meningitis	Fungal Meningitis	Tuberculous Meningitis
WBCs	> 1000 cmm	< 1000 cmm	Variable	
Predominan t differential cell	PMN	Lymphocyte		
Protein	↑↑↑	Normal or ↑	↑	↑
Glucose (better CSF / serum ratio)	↓	Normal	↓	↓

Acute Bacterial Meningitis

Differential Diagnosis

- Febrile seizures
- Tuberculous meningitis
- Viral meningo-encephalitis
- ADEM – acute demyelinating encephalomyelitis
- Brain abscess
- Cerebral malaria
- Intracranial hemorrhage
- Meningismus – neck stiffness due to pneumonia, tonsillitis
- Drugs / Poisoning / Metabolic Encephalopathy

Cerebral Malaria

Clinical Presentation, Diagnosis, Management

- **Etiology** – *Plasmodium falciparum*
- **Pathogenesis** – Hypoxia caused by stasis of microcirculation due to stacking of RBCs infected by malarial parasite
- **Clinical Features** –
 - History – High Fever and coma, Seizures may occur
 - Examination – SOMI usually negative, No other focus of infection
- **CSF examination** – No abnormal cells
- **Blood examination** – No leukocytosis, ICT malaria positive
- **Management** – IV or IM artemether, IV Quinine

Acute Bacterial Meningitis

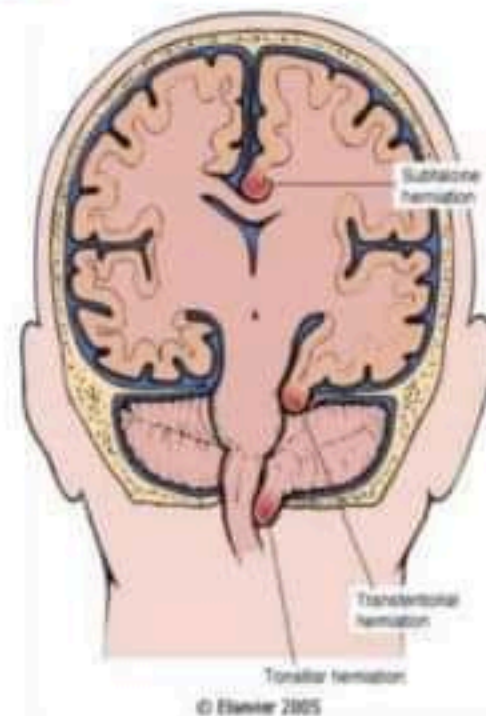
Diagnosis

- **Clinical Features** –
 - History – Fever and seizures
 - Examination – SOMI, No other focus of infection
- **CSF examination** – Polymorphonuclear leukocytes,
 - gram stain positive, bacterial identification positive,
 - bacterial culture positive
- **Blood examination** – leukocytosis, polymorphs increased
- CRP increased
- **Radiology:** CT / MRI Brain – contrast enhancement (increased vascularity)

Acute Bacterial Meningitis

Acute Complications

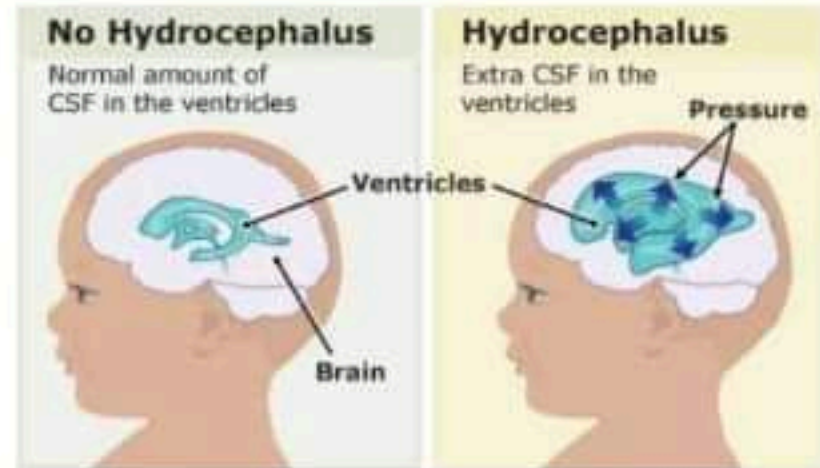
- Raised Intracranial Pressure
- Subdural effusion
- Cerebral edema
- Hemiplegia
- Aspiration pneumonia
- Status epilepticus
- Brain herniation syndromes – tentorial herniation, cerebellar herniation



Acute Bacterial Meningitis

Chronic Complications

- Hemiplegia
- Cranial Nerve Palsies
- Cerebral Palsy
- Intellectual handicap
- Epilepsy
- Deafness
- Hydrocephalus



Acute Bacterial Meningitis

Management

Acute Bacterial Meningitis

Symptomatic Management

- **General Management:**
 - Airway – Breathing – Circulation
 - Monitoring – GCS, Vitals, SaO₂
 - Care of unconscious child
- **Fluids / Feeding:**
 - IV fluids – 0.9 % saline / dextrose-saline
 - Gavage feeding (if needed)
- **Fever:**
 - IV Paracetamol
 - Sponging with water

Acute Bacterial Meningitis

Supportive Management

- **Seizure Management:**

- Acute seizures - IV diazepam or IV midazolam
- Prevent seizures - Phenobarbitone, Leviteracetam, Sodium valproate, Phenytoin

- **Cerebral edema:**

- Restricted IV fluids
- IV Mannitol

- **Reduce inflammation:**

- Dexamethasone – first 48 hours

Acute Bacterial Meningitis

Specific Management

- **Antibiotics** (with CSF penetration):
 - **Initial Therapy:**
 - **Vancomycin**
 - Ceftriaxone / Cefotaxime
 - Meropenem
 - **Therapy after detection of bacteria:**
 - **Gram positive infections – Vancomycin, Ceftriaxone**
 - **Gram negative infections – Ceftriaxone, Meropenem, Amikacin (in infants)**

Acute Bacterial Meningitis

Prognosis

- Age – worse at younger age
- Management – Early and Effective Management of child
- Specific micro-organism

Acute Bacterial Meningitis

Prevention

- **Vaccination**

- Pneumococcal vaccine
- Hib vaccine
- Meningococcal vaccine

- **Increase Immunity**

- Breastfeeding, good nutrition
- Micronutrients

- **Self Protection**

- ❖ Handwashing
- ❖ Social distancing and Face Masks

New EPI Schedule Punjab

Age	EPI Vaccination Schedule (Children)
At Birth	OPV-0, BCG, Hep-B BD
6 Weeks	OPV-1, Rota-1, PCV-1, Penta-1
10 Weeks	OPV-2, Rota-2, PCV-2, Penta-2
14 Weeks	OPV-3, IPV-1, PCV-3, Penta-3
9 Months	TCV, MR-1, IPV-2
15 Months	MR-2
18 Months	DTP Booster

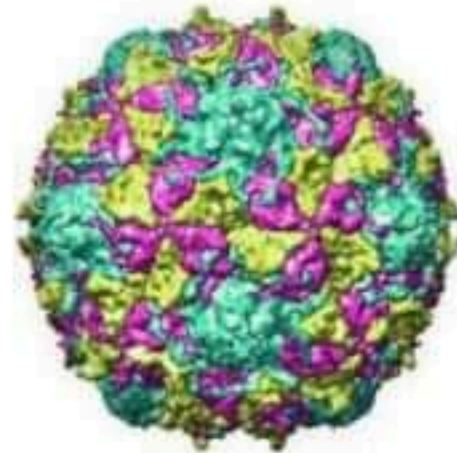


Meningo-encephalitis

Etiology, Clinical Features, Diagnosis, Complications,
Management, Prognosis and Prevention

Meningo-encephalitis

Etiology

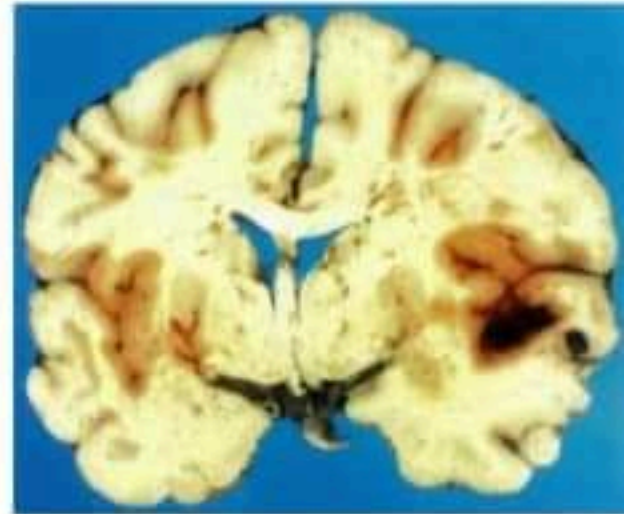


- **Viruses –**
 - Enteroviruses, Polio virus
 - Measles virus, mumps virus
 - Influenza, Parainfluenza virus
 - Varicella virus, Herpes virus,
 - Arbovirus – arthropod transmitted infections
 - Corona virus, Covid 19
- **Protozoa –** Primary Amebic Meningoencephalitis (Naegleria fowleri)
- **Immune mediated –** ADEM (Acute Disseminated Encephalomyelitis), anti-NMDA receptor encephalitis

Meningo-encephalitis

Pathology

- **Meningitis –**
 - Inflammation of meninges
 - SOMI
- **Encephalitis –**
 - Inflammation of Brain
 - Seizures, impaired conscious level
 - Neurological deficits



Meningo-encephalitis

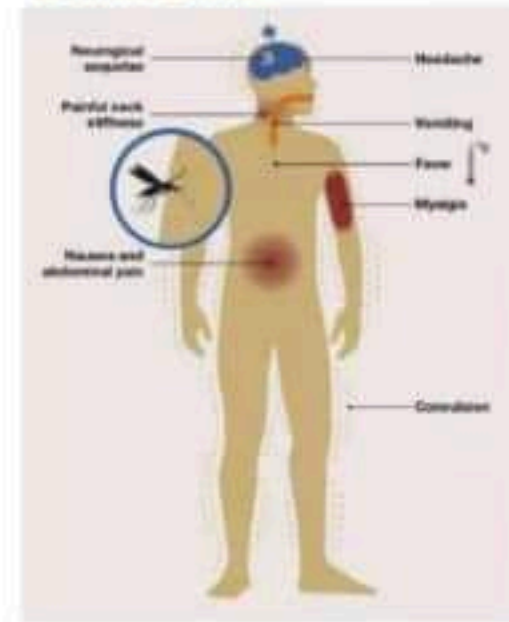
Clinical Features

- **Symptoms**

- Fever – low or high
- Irritability, lethargy
- Headache (in older children), Vomiting
- Seizures / Convulsions, abnormal behavior / movements

- **Signs**

- Impaired conscious level
- SOMI (Neck stiffness, Kernig sign, Brudzinski sign)
- Associated diseases – Measles / Varicella rash, Mumps



Meningo-encephalitis

Evaluation

- **History:**

- Present illness – fever, excessive crying, seizures
- Past History – vaccinations
- Development history – milestones

- **Physical Examination:**

- General Physical Examination – Skin Rash
- Neurological Examination – abnormal movements, SOMI
- Direct Ophthalmoscopy – Papilledema (raised ICP)

Meningo-encephalitis

Investigations

- **CSF examination:**

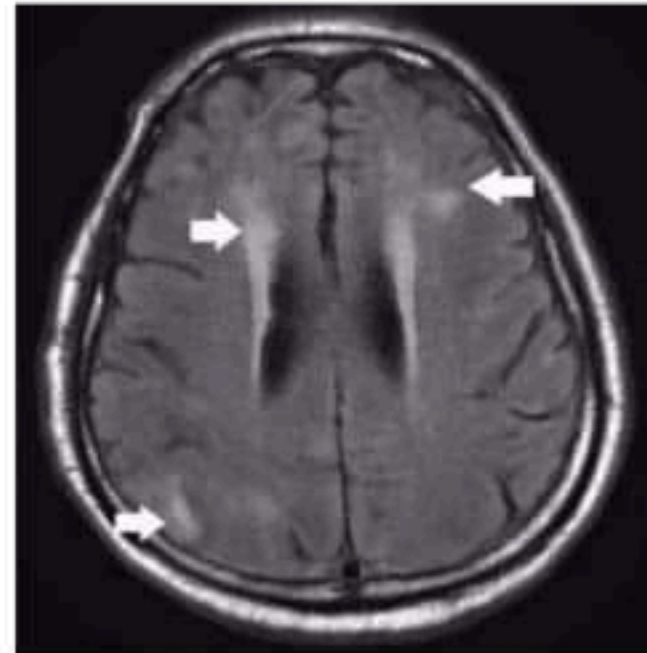
- Biochemistry, Cells
- Virus identification by ELISA / PCR
- Viral isolation
- Specific antibodies

- **Blood:**

- CBC, CRP,
- Viral isolation

- **Radiology:**

- CT / MRI Brain



Meningo-encephalitis

Differential Diagnosis

- Febrile seizures
- Bacterial meningitis
- Cerebral malaria
- Intracranial hemorrhage
- ADEM – acute demyelinating encephalomyelitis
- Meningismus – neck stiffness due to pneumonia, tonsillitis
- Drugs / Poisoning / Metabolic Encephalopathy

Meningo-encephalitis

Investigations – CSF analysis

CSF analysis in different types of meningitis

	Bact. Meningitis	Viral Meningitis	Fungal Meningitis	Tuberculous Meningitis
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Predominan t differential cell	PMN	Lymphocyte		
Protein	↑↑↑	Normal or ↑	↑	↑
Glucose (better CSF / serum ratio)	↓	Normal	↓	↓

Meningo-encephalitis

Diagnosis

- **Clinical Features** –
 - History – Fever, Seizures, Vomiting, Abnormal movements
 - Examination – SOMI may be negative, Skin rash
- **CSF examination** – Lymphocytes, Viral isolation
- **Blood examination** – leukopenia, Viral isolation
- **Radiology** – CT / MRI Brain – contrast enhancement (increased vascularity)

Meningo-encephalitis

Complications

- Raised Intracranial Pressure
- Cerebral edema
- Cranial Nerve Palsies
- Brain herniation syndromes –
tentorial herniation, cerebellar herniation
- Cerebral Palsy / Intellectual handicap
- Epilepsy
- Movement disorders



Meningo-encephalitis

Management

Meningo-encephalitis

Supportive Management

- **Seizure Management:**

- Acute seizures - IV diazepam or IV midazolam
- Prevent seizures - Phenobarbitone, Leviteracetam, Sodium valproate, Phenytoin

- **Raised ICP:**

- IV Mannitol

Meningo-encephalitis

Specific Management

- **Anti – Viral Therapy:**
 - Acyclovir – Herpes encephalitis, Varicella encephalitis
- **Immune-mediated:**
 - Corticosteroids – IV Methylprednisolone

Meningo-encephalitis Prevention

- **Vaccination**

- Measles vaccine
- Mumps vaccine
- Polio vaccine
- Varicella vaccine
- Covid vaccine

- **Increase Immunity**

- Breastfeeding, good nutrition, **Micronutrients**
- Handwashing, **Social distancing**

New EPI Schedule Punjab	
Age	EPI Vaccination Schedule (Children)
At Birth	OPV-0, BCG, Hep-B BD
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15 Months	MR-2
18 Months	DTP Booster

SSPE – Subacute Sclerosing Panencephalitis

Clinical Presentation, Diagnosis, Management

- **Epidemiology** – seen in older children
- **Etiology** – Previous Measles infection
- **Pathogenesis** – activation of measles virus causing encephalitis
- **Clinical Features** – Myoclonic seizures
- **EEG** – burst – suppression pattern
- **CSF examination** – No abnormal cells, measles antibodies
- **Management** – Symptomatic
- **Prognosis** – Poor